

Tutorial 6 Group Work

Suppose now that the true proportion p for type A modified plants is in fact 0.519. Then, a randomly sampled type A plant will be susceptible with probability 0.519. 100 samples each of size 104 are obtained from type A plants, and they are exposed to the disease once more. The data is recorded in `104_plants.csv` which has 100 rows and 105 columns. Each row of it is for one random sample which has the information on the susceptibility of 104 plants of type A to a specific disease. The first column is the sample ID.

- (a) Import the given dataset into R. Write the code such that for each sample we can calculate (and save) the proportion of susceptible for each sample and a 95% CI for p . **For the first sample** which is the first row of the data, try to figure out (1) how to use R to find the proportion of plants that is susceptible to the disease and (2) how to use R to form a 95% CI for the proportion of plants that are susceptible to the disease. Lastly, try to figure out how to do part (1) and (2) for all the 100 samples in the data.

Hint: Using `for` loop and `if` functions in R can help very much in writing effective and short codes for this question.

- (b) Among the 100 confidence intervals obtained above, how many of them cover 0.519? Why is it not exactly 95?
- (c) Plot a histogram for the 100 sample proportions obtained in part (a). Does the histogram resemble a bell-shaped curve?
- (d) Derive the mean and standard deviation of all the 100 sample proportions obtained in part (a). Are the mean and standard deviation close to 0.519 and $\sqrt{0.519(1 - 0.519)/104} = 0.05$ respectively? Why?